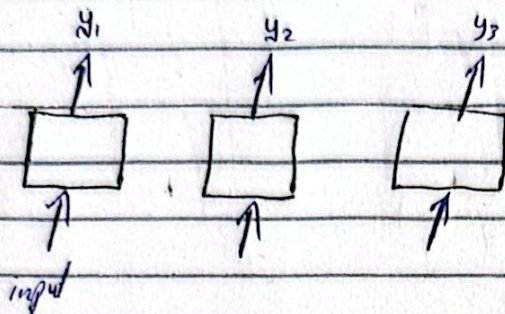


⇒ Encoder & Decoder:

⇒ Many To Many RNN.



* Langs Translation
Eng → French

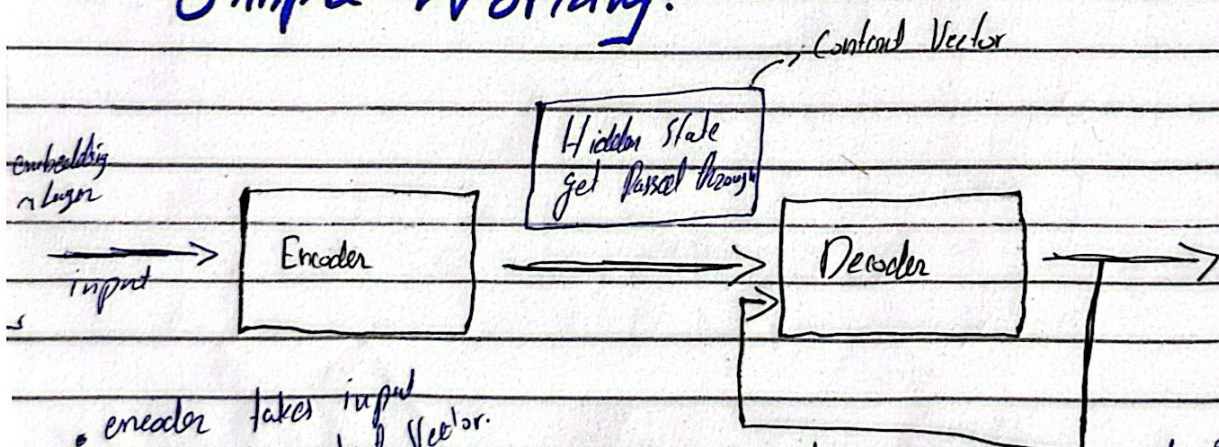
* Also called "Sequence to Sequence" Neural Network.

⇒ Input is Sequence of Data
⇒ Output is Sequence of Data.

AK In this case when "I/P & O/P" both are sequential data we can not use RNN, LSTM, GRU etc.

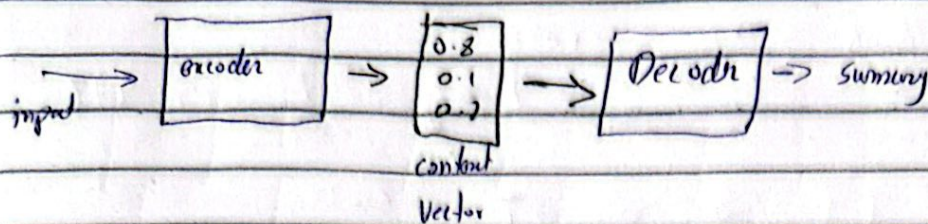
AA We Use Encoder Decoder.

↳ Simple Working:



• encoder takes input & generate context vector.

• Decoder generates output word-by-word & keeps feeding the previous words into the decoder again.



↳ 1: Encoder: Take I/P & convert it into Context Vector

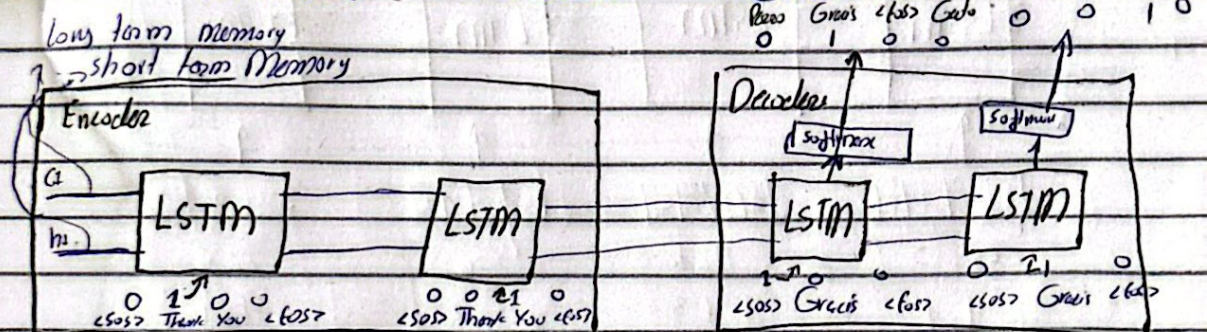
↳ 2: Decoder: Takes Context Vector & generate output.

⇒ Use - Cases:

Language Translation
Text generation
Text Suggestion.

→ Sequence To Sequence Encoder

Decoder Neural Network:



Dataset:

English

French

<SOS> Thank You <EOS>

<SOS> Gracis <EOS>

• along with sentence add <SOS> & <EOS>

→ SOS ⇒ Start of statement

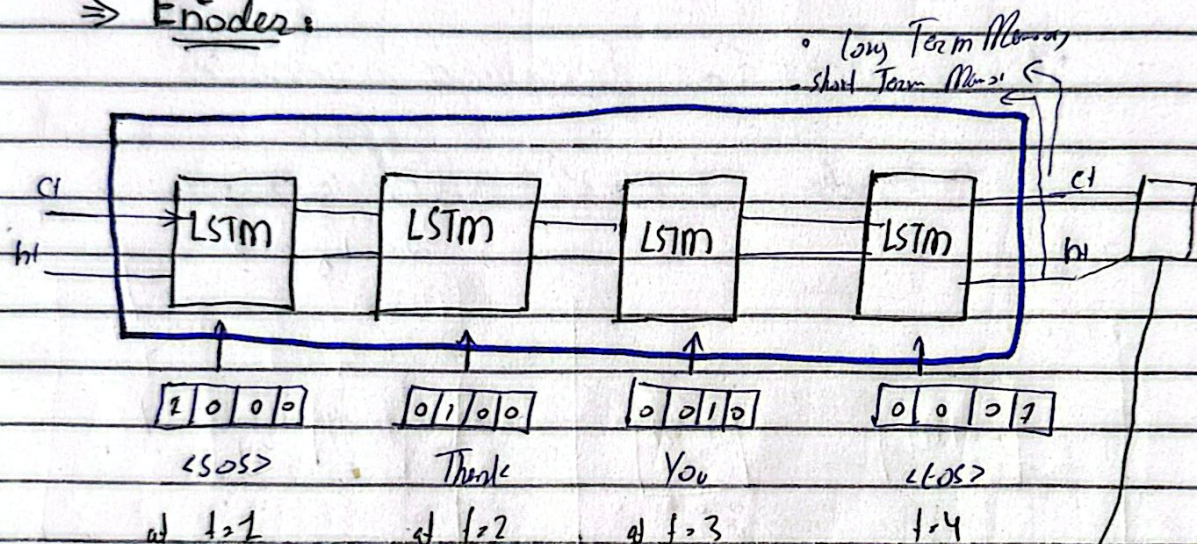
→ EOS ⇒ End of "

$\langle sos \rangle$ Thank You $\langle eos \rangle$
 $[- \quad - \quad - \quad -] \Rightarrow$ if we are using OHE

$t=1 \Rightarrow [1 \quad 0 \quad 0 \quad 0] \quad \langle sos \rangle$
 $t=2 \Rightarrow [0 \quad 1 \quad 0 \quad 0] \quad \text{Thank}$
 $t=3 \Rightarrow [0 \quad 0 \quad 1 \quad 0] \quad \text{You}$
 $t=4 \Rightarrow [0 \quad 0 \quad 0 \quad 1] \quad \langle eos \rangle$

* As we know we pass a sentence word by word to LSTM at different time steps.

$\Rightarrow$ Encoder:

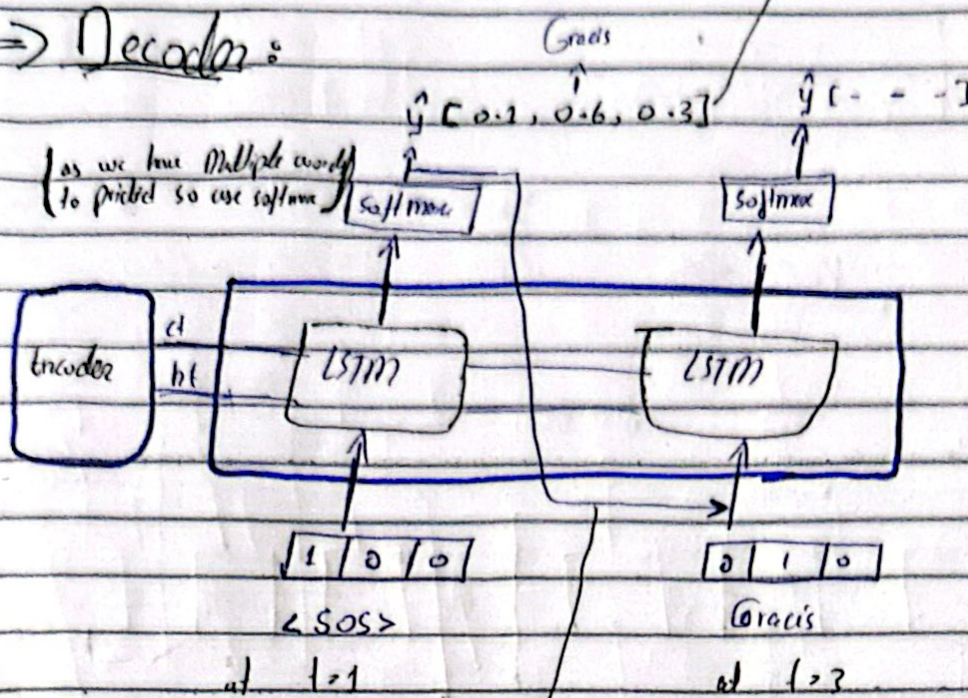


* This ci & hi is called Context Vector & is passed to next stage.

that is Decoder.

Training data $\Rightarrow$ $\langle \text{SOS} \rangle$ Gracis $\langle \text{FOS} \rangle$

$\Rightarrow$ Decoder:



$\rightarrow$ Pass output To next LSTM.

$\rightarrow$ at $t=1$ we got $\hat{y}$ & we have y

$\hat{y} [- - -]$ (Predicted Value)

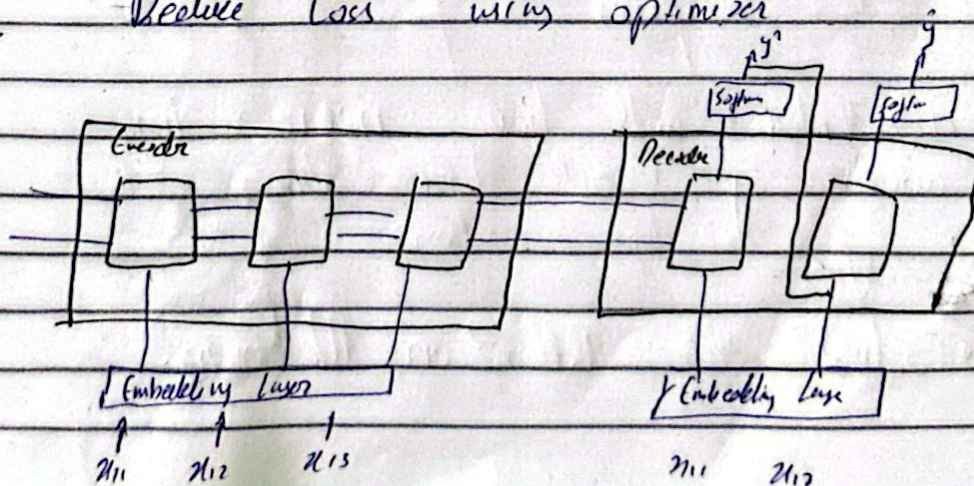
$y [- - -]$ (Actual Value)

$\rightarrow$ at $t=2$, we got $\hat{y}$ & we have y

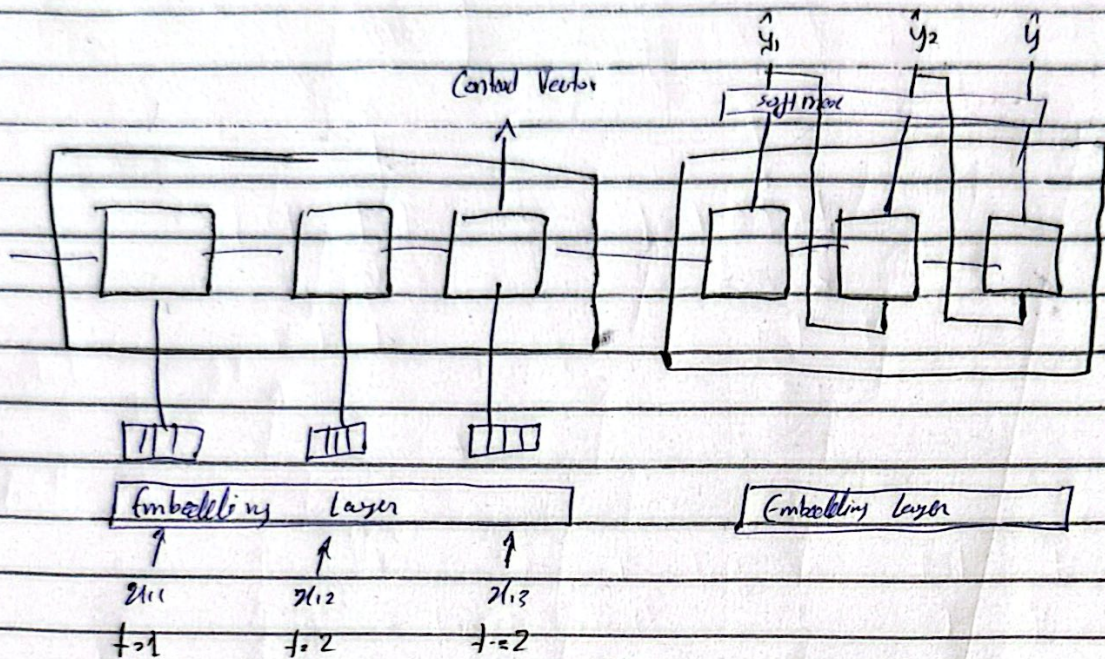
$\hat{y} [- - -]$

$y [- - -]$

Reduce Loss using optimizer



⇒ Problems with Encoder & Decoder.



★ Context Vector Represents The Entire Sentence.

⇒ As Sentence length increases the Accuracy started to decreasing

⇒ Context Vector will have more info of words of recent time stamps but not for time stamps.

• Solved via Attention Mechanisms.